

1. Business Understanding:

The business problem revolves around understanding sales performance, profitability, customer behaviour, product demand, and regional trends within a retail superstore.

The organization wants to identify:

- Which product categories generate the highest revenue
- Which regions contribute the most profit
- Impact of discounts on profitability
- Customer purchasing patterns
- Monthly sales trends
- Best-performing products
- Overall business performance

The goal is to help the business make data-driven decisions regarding pricing strategies, inventory management, marketing campaigns, customer targeting, and revenue optimization.

2. Dataset Overview:

The dataset contains transactional sales data from a retail superstore.

Column	Description
Order ID	Unique order identifier
Order Date	Date of purchase
Ship Date	Date of shipment
Customer Name	Customer details
Region	Sales region
Category	Product category
Sales	Revenue generated
Quantity	Units sold
Discount	Discount applied
Profit	Profit earned

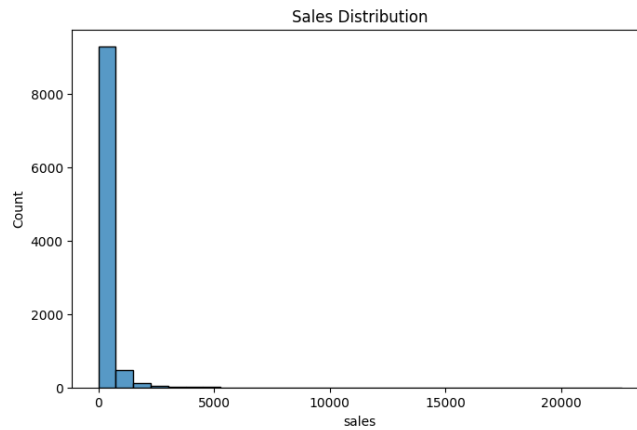
Created Columns

- $\text{Delivery_Days} = \text{Ship Date} - \text{Order Date}$
- $\text{Profit_Category} = \text{Profit} / \text{Loss}$
- $\text{Sales_Category} = \text{Low} / \text{Medium} / \text{High}$
- $\text{Month} = \text{Month extracted from Order Date}$

3. Univariate Analysis:

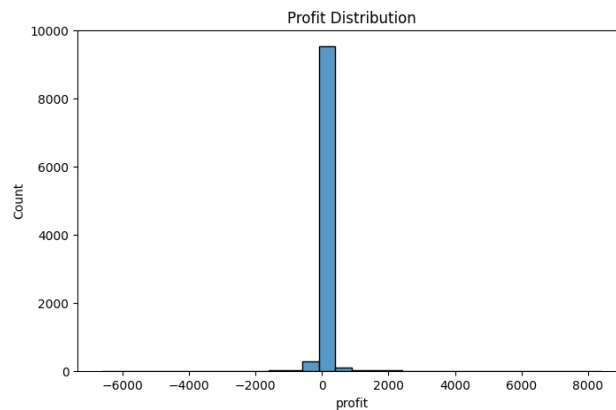
3.1 Sales Distribution

- Sales distribution is highly right-skewed.
- Most transactions involve small sales amounts.
- Few transactions generate extremely high revenue.



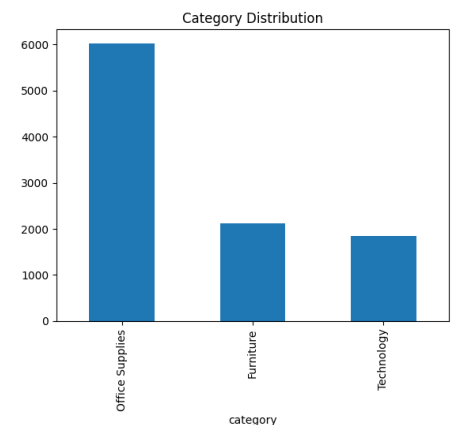
3.2 Profit Distribution

- Majority of transactions generate positive profit.
- Some transactions result in losses.
- Profit variability is high across products.



3.3 Category Distribution

- Office Supplies contains the highest number of products.
- Technology products contribute significant revenue.



3.4 Summary Statistics

	row_id	order_date	postal_code	sales	quantity	discount	profit	delivery_days	month
count	9994.000000	9994	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	2016-04-30 00:07:12.259355648	55190.379428	229.858001	3.789574	0.156203	28.656896	3.958175	7.809686
min	1.000000	2014-01-03 00:00:00	1040.000000	0.444000	1.000000	0.000000	-6599.978000	0.000000	1.000000
25%	2499.250000	2015-05-23 00:00:00	23223.000000	17.280000	2.000000	0.000000	1.728750	3.000000	5.000000
50%	4997.500000	2016-06-26 00:00:00	56430.500000	54.490000	3.000000	0.200000	8.666500	4.000000	9.000000
75%	7495.750000	2017-05-14 00:00:00	90008.000000	209.940000	5.000000	0.200000	29.364000	5.000000	11.000000
max	9994.000000	2017-12-30 00:00:00	99301.000000	22638.480000	14.000000	0.800000	8399.976000	7.000000	12.000000
std	2885.163629	NaN	32063.693350	623.245101	2.225110	0.206452	234.260108	1.747567	3.284654

4. SQL Business Questions:

1. Top 5 Products by Revenue -

```
SELECT product_name,
SUM(sales) AS revenue
FROM df
GROUP BY product_name
ORDER BY revenue DESC
LIMIT 5;
```

	product_name	revenue
0	Canon imageCLASS 2200 Advanced Copier	61599.824
1	Fellowes PB500 Electric Punch Plastic Comb Bin...	27453.384
2	Cisco TelePresence System EX90 Videoconferenci...	22638.480
3	HON 5400 Series Task Chairs for Big and Tall	21870.576
4	GBC DocuBind TL300 Electric Binding System	19823.479

2. Top 5 Customers by Revenue –

```
SELECT customer_name,
SUM(sales) AS revenue
FROM df
GROUP BY customer_name
ORDER BY revenue DESC
LIMIT 5;
```

	customer_name	revenue
0	Sean Miller	25043.050
1	Tamara Chand	19052.218
2	Raymond Buch	15117.339
3	Tom Ashbrook	14595.620
4	Adrian Barton	14473.571

3. Region-wise Profit –

```
SELECT region,  
SUM(profit) AS total_profit  
FROM df  
GROUP BY region  
ORDER BY total_profit DESC;
```

	region	total_profit
0	West	108418.4489
1	East	91522.7800
2	South	46749.4303
3	Central	39706.3625

4. Category-wise Revenue –

```
SELECT category,  
SUM(sales) AS revenue  
FROM df  
GROUP BY category;
```

	category	revenue
0	Furniture	741999.7953
1	Office Supplies	719047.0320
2	Technology	836154.0330

5. Average Discount –

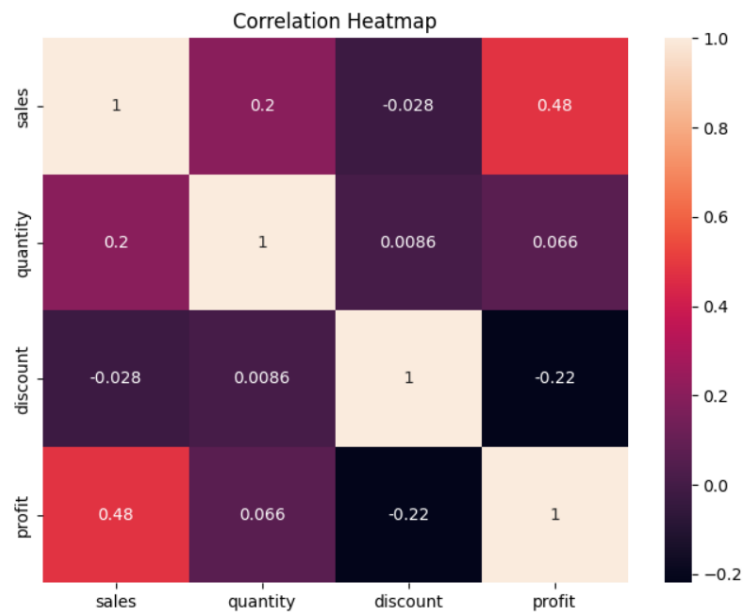
```
SELECT AVG(discount)  
FROM df;
```

	AVG(discount)
0	0.156203

5. Multivariate Analysis:

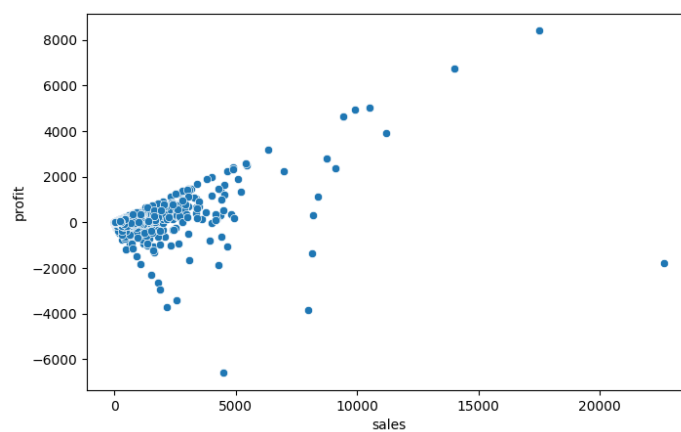
Correlation Matrix

- Sales and Profit show positive correlation.
- Discount has negative correlation with Profit.
- Quantity has weak positive relationship with Sales.
- Higher discounts often reduce profitability.



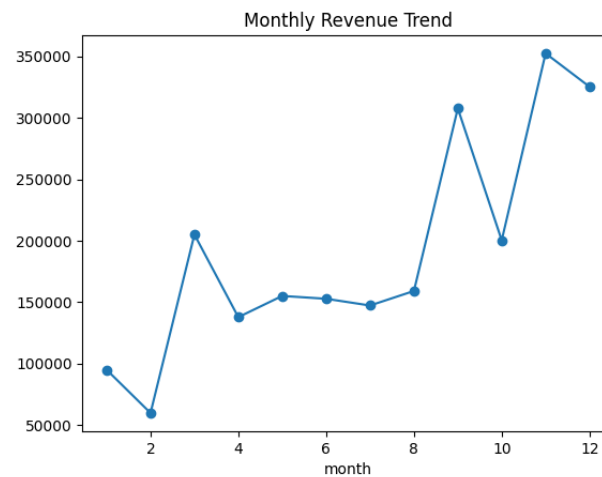
Sales vs Profit

- Higher sales generally lead to higher profit.
- Several high-sales transactions still generate losses.



Monthly Revenue Trend

- Revenue fluctuates across months.
- Certain months show significantly higher sales.



6. Dashboard KPIs:

KPI	Value
Total Sales	\$2.30M
Total Profit	\$286.40K
Total Orders	5,009
Average Discount	15.6%

7. Key Business Insights:

- West Region generated the highest profit.
- Technology category generated the highest revenue.
- Discounts negatively impacted profitability.
- A small number of products contributed significantly to overall sales.
- Customer purchasing patterns vary across regions.
- Sales and Profit exhibit a positive relationship.

8. Conclusion:

The analysis successfully identified important business patterns within the Superstore dataset.

The findings indicate that product category, regional performance, discount strategies, and customer purchasing behavior significantly influence revenue and profitability.

These insights can help decision-makers improve pricing policies, inventory planning, customer targeting, and overall business growth.